

When Small Action Errors Matter: Closed-Loop Analysis of Post-Training Quantization for VLA Policies

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Abstract

Post-training quantization (PTQ) is usually presented as a compression procedure: a full-precision model is mapped into a low-precision representation, and the resulting model is evaluated by output drift, task accuracy, or deployment efficiency. For vision-language-action (VLA) policies, this framing is incomplete. A VLA model is not only a predictor; it is a closed-loop controller. We argue that PTQ for VLA policies should be evaluated as a closed-loop policy perturbation. Quantization changes the implemented policy function, which can change the state distribution induced by robot-environment interaction. Small action errors can therefore be amplified by contact dynamics, receding-horizon feedback, and thresholded success conditions.

We study this effect in a reproduction-oriented analysis of selective W4A8 fake quantization for a GR00T N1.5 policy on LIBERO-10. Offline teacher-student action drift shows that attention balancing methods can improve mean action error, but also worsen a non-trivial subset of held-out observations. In closed-loop simulation over 150 matched task-initialization pairs, selective W4A8 variants remain in the FP16 behavioral range under our evaluation protocol, but their gains are not monotonic: compensation methods repair some failed rollouts while introducing new failures elsewhere. The highest point-estimate single compensation mode, output head balancing (OHB), reaches 116/150 successes compared with 113/150 for uncompensated W4A8 and 108/150 for FP16, yet it still regresses several task slices. We do not claim statistically significant closed-loop superiority from these small aggregate gaps. Our study focuses on behavior under fake quantization rather than packed-kernel deployment efficiency. The results suggest that VLA quantization should be evaluated as policy perturbation, not merely as numerical approximation.

1 Introduction

Large vision-language-action models are increasingly used as generalist robot policies. They map visual observations and language instructions into action chunks, often through diffusion-style denoising or transformer action heads. As these models become larger, quantization is a natural route toward cheaper deployment. However, common quantization evaluations focus on local approximation quality: weight error, activation error, output MSE, or single-step action drift. These metrics are useful, but they do not fully capture what matters for embodied control.

In a closed-loop policy, an action error at time t changes the next observation. The next action is then produced on a state that may not have been visited by the original full-precision policy. This feedback loop creates a mismatch between local teacher-student error and final task success. The issue is especially sharp in manipulation, where small geometric deviations can change contact timing, grasp stability, object pose, or whether a state satisfies a binary success predicate.

This paper studies the following question:

When a VLA policy is post-training quantized, how should we interpret small action errors in terms of closed-loop task success?

Our answer is that PTQ should be treated as a policy perturbation. It maps a full-precision policy into a constrained low-precision function class, thereby changing both the conditional action distribution and the

induced state distribution. The relevant empirical object is not only local action drift on fixed observations, but also how the perturbed policy redistributes success and failure across matched closed-loop rollouts.

We make three observations:

1. Selective W4A8 quantization can be behaviorally robust even when it measurably changes action outputs.
2. Attention-balancing corrections can improve mean offline drift while still worsening individual observations and task slices.
3. Paired closed-loop rollout flips expose repair/regression structure that aggregate success rates hide.

Our contributions are:

1. We formulate PTQ for VLA policies as a closed-loop policy perturbation problem, where the main object is the induced state distribution rather than only local action approximation.
2. We provide a reproduction-oriented GR00T/LIBERO case study showing that selective W4A8 fake quantization can remain in the FP16 behavioral range while changing which task-initialization pairs succeed.
3. We show that offline mean drift and aggregate success rate can both obscure important structure, and advocate paired repair/regression analysis as a compact diagnostic for VLA quantization.

Scope of claims. This paper should be read as a behavior-level reproduction study and diagnostic analysis, not as a new quantization algorithm or a deployment benchmark. We make no latency, memory, or energy claim because our implementation uses fake quantization rather than packed low-precision kernels. We also do not treat small aggregate success-rate differences as evidence of policy dominance. The main claim is narrower: VLA quantization can preserve aggregate success while redistributing which closed-loop rollouts succeed, so paired rollout analysis is necessary to interpret the effect of small action perturbations.

2 Related Work

Vision-language-action policies. Recent robot policies increasingly combine large-scale visual, language, and action modeling. RT-1 introduced a scalable transformer policy trained on diverse real-robot data [2]. RT-2 framed robotic actions as tokens inside a vision-language-action model and showed transfer from web-scale vision-language training to control [3]. OpenVLA made this direction more accessible by releasing a 7B-parameter open-source VLA trained on large robot demonstration mixtures, and also noted practical interest in efficient fine-tuning and quantized serving [7]. GR00T N1 uses a dual-system VLA architecture where a vision-language module conditions a diffusion transformer action module [10]. Diffusion Policy and related action-diffusion methods model robot behavior as conditional denoising, often with receding-horizon control [4]. Our work focuses on the evaluation problem that appears when such policies are compressed after training.

Post-training quantization. PTQ methods for large transformers often target memory and latency while preserving model quality. LLM.int8() handles transformer outliers through mixed-precision decomposition [5]; GPTQ uses approximate second-order information for one-shot low-bit weight quantization [6]; SmoothQuant migrates quantization difficulty from activations to weights through an equivalent smoothing transform [12]; and AWQ uses activation statistics to protect salient weight channels [8]. These methods demonstrate that transformer quantization can be accurate when calibration accounts for activation structure. However, most evaluations are open-loop language, vision-language, or recognition tasks. In VLA policies, the quantized model selects actions that alter future inputs, so the calibration and evaluation target must include closed-loop effects.

Quantization for VLA models. QuantVLA is, to our knowledge, the most direct prior work on PTQ for VLA systems. It proposes a scale-calibrated PTQ framework with selective quantization, attention temperature matching, and output head balancing, and reports memory and latency gains together with LIBERO success-rate improvements [13]. Our study is not a deployment-efficiency reproduction of those claims. Instead, we use a behavior-level fake-quantization reproduction to analyze why VLA quantization can improve some rollouts while regressing others, and why paired closed-loop evaluation is necessary even when average offline drift improves.

Sequential prediction and distribution shift. The gap between offline error and closed-loop behavior is a classic issue in imitation learning and sequential prediction. Dagger formalizes the problem that future observations depend on previous actions, so policies trained or evaluated only under a fixed data distribution can suffer from compounding errors [11]. VLA quantization creates a related but distinct problem: the policy is already trained, but PTQ perturbs its action outputs and therefore perturbs the state distribution it induces. This makes paired rollout analysis a natural complement to offline teacher-student action drift.

Feedback control perspective. Classical feedback control emphasizes that closed-loop behavior is determined by the interaction between controller perturbations, plant dynamics, and feedback gain [1]. We use this lens to interpret PTQ not as a static approximation problem, but as a bounded policy perturbation whose effect depends on local closed-loop sensitivity and task margins.

3 Quantization as Policy Perturbation

Let $\pi_\theta(a \mid s, l)$ denote a full-precision VLA policy conditioned on state observation s and language instruction l . PTQ constructs a quantized policy

$$\theta_q = Q(\theta; C), \quad \pi_q = \pi_{\theta_q}, \quad (1)$$

where Q is a calibration-dependent projection into a low-precision parameterization and C is a calibration set. Even without gradient updates, this is not a neutral storage transform: it changes the implemented function.

Offline action drift measures a local difference such as

$$e(s) = \|\pi_q(s, l) - \pi_\theta(s, l)\| \quad (2)$$

on observations sampled from some dataset or teacher rollout distribution. Closed-loop performance depends on the state distributions induced by the two policies, d_{π_θ} and d_{π_q} .

For a local transition model,

$$s_{t+1} = f(s_t, a_t), \quad a_t = \pi(s_t), \quad (3)$$

a first-order error expansion gives

$$\delta s_{t+1} \approx J_s \delta s_t + J_a \delta a_t, \quad \delta a_t = \pi_q(s_t) - \pi_\theta(s_t). \quad (4)$$

Even if δa_t is small on the teacher state distribution, the accumulated δs_t can move the quantized policy into states where the original local error estimate no longer applies.

This is why manipulation success is not a smooth function of single-step action MSE. A rollout can flip because of a small action perturbation if that perturbation changes whether the gripper reaches the object before closing, whether a placed object crosses a success-region boundary, whether an object contact produces a different pose, or whether a receding-horizon action chunk enters a different trajectory basin. In this view, quantization is closer to post-training policy editing than to passive compression.

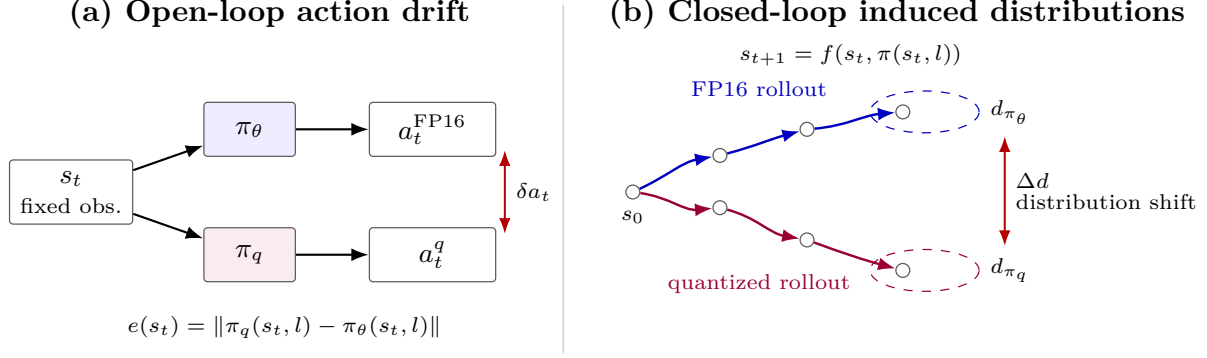


Figure 1: Quantization as closed-loop policy perturbation. Offline action drift compares two actions on a fixed observation, while closed-loop execution lets each policy induce its own future observation distribution.

4 Experimental Setup

Model and task. We use the GR00T N1.5 LIBERO long-horizon checkpoint and evaluate on LIBERO-10 [9, 10]. The accepted FP16 baseline matches the official reference result of 38/50 on the standard 5-trial LIBERO-10 run. The evaluated policy predicts action chunks with seven action components: x , y , z , roll, pitch, yaw, and gripper. Unless noted otherwise, experiments use 8 denoising steps.

Quantization scope. We study the selective `llm_dit_mlp` W4A8 fake-quantization scope: 84 selected LLM linear layers and 32 selected DiT MLP linear layers, for 116 quantized modules. DiT attention projections are intentionally left in floating point. This design matters because attention projections feed a softmax routing mechanism, while MLP errors are more often residual perturbations.

Compensation modes. Following QuantVLA terminology, we evaluate two scale-calibrated compensation mechanisms: attention temperature matching (ATM) and output head balancing (OHB) [13]. Our implementation should be read as a reproduction-oriented approximation of these mechanisms rather than a claim of exact code-level equivalence to the original implementation. ATM rescales the DiT attention query to match teacher/student attention-logit standard deviation:

$$\alpha = \frac{\text{std}_{\text{teacher}}(\text{attention logits})}{\text{std}_{\text{student}}(\text{attention logits})}, \quad \text{query} \leftarrow \alpha \cdot \text{query}. \quad (5)$$

OHB rescales the attention output before residual addition to match teacher/student attention-output RMS:

$$\beta = \frac{\text{rms}_{\text{teacher}}(\text{attention output})}{\text{rms}_{\text{student}}(\text{attention output})}, \quad \text{attention output} \leftarrow \beta \cdot \text{attention output}. \quad (6)$$

The combined mode applies both corrections. In all cases the compensation acts on DiT attention processors; it does not mean DiT attention weights are quantized.

Offline drift protocol. The offline validation uses 16 real LIBERO calibration observations and 128 held-out real evaluation observations. Teacher and student calls use matched random seeds because action denoising begins from random Gaussian actions. We report NMSE, relative RMSE, cosine similarity, and max absolute difference.

Closed-loop protocol. The main closed-loop analysis evaluates LIBERO-10 initial states 0–14: 10 tasks, 15 initial states per task, and 150 episodes per policy/mode. For paired comparisons, each mode is evaluated on the same task-initialization pairs. We report Wilson 95% confidence intervals for aggregate success rates as a descriptive uncertainty estimate, but use paired repair/regression counts as the primary behavioral diagnostic.

5 Results

5.1 Offline Mean Drift Improves, But Individual Regressions Remain

On the 128 held-out real observations, ATM and OHB reduce mean action drift relative to uncompensated W4A8 (Table 1). The identity control exactly matches `none`, showing that the custom attention processor path itself does not introduce measurable drift.

Table 1: Offline teacher-student action drift on 128 held-out real LIBERO observations.

Config	Mode	Modules	NMSE mean	Rel. RMSE mean	Cosine mean
llm_dit_mlp	none	116	0.0178962	0.0981458	0.992492
llm_dit_mlp	identity	116	0.0178962	0.0981458	0.992492
llm_dit_mlp	ATM	116	0.0159471	0.0904183	0.993101
llm_dit_mlp	OHB	116	0.0160919	0.0853958	0.992769
llm_dit_mlp	ATM+OHB	116	0.0153168	0.0838784	0.993048

The mean improvement hides observation-level regressions. ATM+OHB has the best mean drift, while still worsening 34 of 128 held-out observations by NMSE. This is the first sign that mean offline drift is insufficient: compensation moves the action distribution closer on average, but not uniformly.

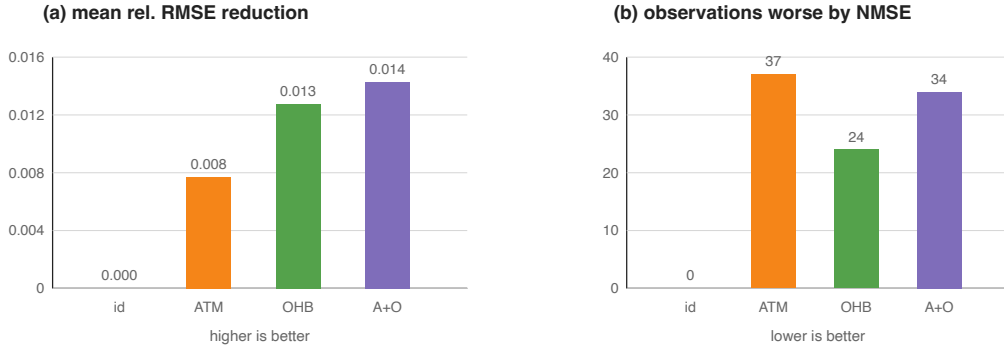


Figure 2: Offline mean drift versus observation-level regressions. Left: reduction in mean relative RMSE from uncompensated W4A8. Right: number of held-out observations whose NMSE worsens relative to uncompensated W4A8.

5.2 Selective W4A8 Remains in the FP16 Range

Over 150 LIBERO-10 task-initialization pairs, selective W4A8 remains in the same performance band as FP16 (Table 2). Some quantized variants have higher aggregate point estimates, but these gaps should not be read as evidence that quantization is inherently better than FP16. Rather, the quantized policies perturb the trajectory distribution. In finite closed-loop evaluation, that perturbation can push some failed FP16 trajectories into successful basins while pushing other successful trajectories into failure.

Table 2: Closed-loop LIBERO-10 success over task-initialization pairs 0–14.

Policy	Successes	Success rate (Wilson 95% CI)
FP16	108/150	72.0% [64.3, 78.6]
W4A8 llm_dit_mlp + none	113/150	75.3% [67.9, 81.5]
W4A8 llm_dit_mlp + ATM	114/150	76.0% [68.6, 82.1]
W4A8 llm_dit_mlp + OHB	116/150	77.3% [70.0, 83.3]
W4A8 llm_dit_mlp + ATM+OHB	114/150	76.0% [68.6, 82.1]

The confidence intervals are wide and overlapping, so the table should not be read as a statistical ranking of policies. The more stable conclusion is that selective W4A8 remains in the same behavioral range as FP16 under this protocol, while changing the identity of successful and failed rollouts.

5.3 Aggregate Rates Hide Task-Level Redistribution

The effects are task-dependent. ATM improves task 4 by 5 successes over **none**, but regresses task 8 by 5 successes. OHB is more balanced and gives the highest aggregate point estimate, but it still regresses tasks 3, 8, and 9 relative to **none**. The detailed task-level table appears in Appendix A.

5.4 Paired Rollout Flips Reveal The Mechanism

Paired comparisons over identical task-initialization pairs show that the compensation modes are not monotonic improvements (Table 3). OHB is not simply “3 episodes better” than **none**; it repairs 16 failures and introduces 13 new failures. Its net gain is small, but the underlying trajectory redistribution is substantial.

Table 3: Paired outcomes over the same 150 task-initialization pairs.

Comparison	Repaired	Regressed	Same success	Same failure	Net
ATM vs none	14	13	100	23	+1
OHB vs none	16	13	100	21	+3
OHB vs ATM	15	13	101	21	+2

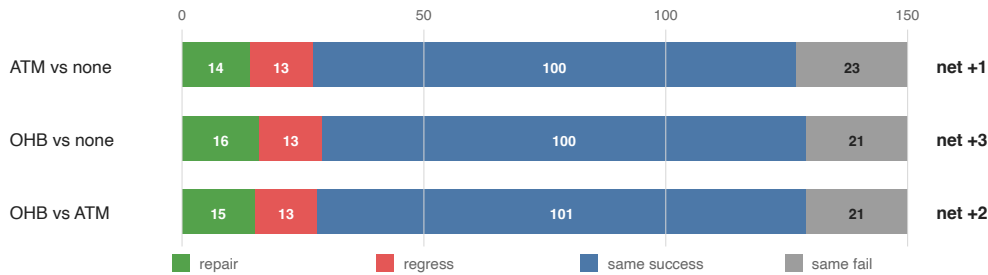


Figure 3: Paired closed-loop rollout flips. Net success-rate changes decompose into repaired failures, new regressions, unchanged successes, and unchanged failures.

For the disjoint init 5–14 comparison between FP16 and ATM+OHB W4A8, we observe the same pattern: 14 FP16 failures are repaired, while 8 FP16 successes become quantized failures. The aggregate gain of 6/100 comes from this difference.

6 Analysis

Why can a quantized policy improve a rollout? The quantized policy is not guaranteed to be worse than FP16 in a finite simulator benchmark. FP16 is not an oracle; it is a learned policy with its own fragile regions. A small action perturbation can move a trajectory away from a failure mode. This is especially plausible when the FP16 policy is already weak on a task slice, such as LIBERO task 8 in our baseline. Therefore, an aggregate improvement should not be read as “low precision is better.” It means the perturbed policy enters different trajectory basins, some of which are more favorable under the benchmark’s finite initial states and success predicates.

Why does offline drift fail to predict closed-loop behavior? Offline drift is measured on a fixed observation set. Closed-loop rollouts produce observations adaptively. Once a quantized action changes the environment state, later observations may differ from the offline distribution. This creates two mismatches: the local action error estimate may not apply to the new states, and the success function may be discontinuous around contact and placement boundaries. This explains why ATM+OHB can have the best mean offline NMSE but not the best closed-loop aggregate success.

A control-theoretic interpretation. From a feedback-control perspective, quantization acts like a structured input disturbance injected into the learned controller. Write

$$a_t^q = \pi_\theta(s_t^q, l) + \eta(s_t^q, l), \quad (7)$$

where η is the action perturbation induced by quantization and compensation. Linearizing around an FP16 trajectory gives

$$\delta s_{t+1} \approx (A_t + B_t K_t) \delta s_t + B_t \eta_t, \quad (8)$$

where $A_t = \partial f / \partial s$, $B_t = \partial f / \partial a$, and $K_t = \partial \pi_\theta / \partial s$ along the trajectory. The closed-loop sensitivity is therefore governed not only by the magnitude of η_t , but also by the local closed-loop gain induced by the environment dynamics and the policy feedback. A small action perturbation can be attenuated in low-gain regions and amplified near contact, grasping, or placement transitions.

This also makes success a margin problem. Let $h(s_T)$ denote an implicit terminal success margin, with success when $h(s_T) > 0$. A rollout with large positive margin can tolerate sizeable action perturbations, while a low-margin rollout can flip if quantization moves $h(s_T)$ across zero. This explains why paired outcomes contain both repairs and regressions: quantization can move a trajectory away from one failure basin while pushing another trajectory across a success boundary. The useful diagnostic is therefore not just average action error, but the alignment between action perturbations, closed-loop sensitivity, and task success margins.

Why is selective W4A8 robust? The `llm_dit_mlp` scope leaves DiT attention projections in floating point. This avoids quantizing the most routing-sensitive part of the action head. A perturbation to $QK^\top$ before softmax can change attention entropy and token routing. By contrast, MLP quantization more often enters as an additive residual perturbation,

$$y_{\text{student}} = y_{\text{teacher}} + \epsilon. \quad (9)$$

Layer normalization, residual connections, diffusion denoising, and receding-horizon replanning can absorb some of this error. This helps explain why uncompensated W4A8 already reaches 113/150 successes.

Why are ATM and OHB not additive? ATM changes where attention looks by changing the attention-logit scale. OHB changes how strongly the attention output enters the residual stream. If ATM moves routing into a different pattern, OHB rescales a different output than the one calibrated under the uncompensated student. The combined correction can overcorrect, undercorrect, or move the policy into a different trajectory basin. This is consistent with ATM+OHB reaching 114/150, below standalone OHB at 116/150.

What should be reported for VLA quantization? The minimum evidence package should include offline action drift on held-out real observations, per-component action error, closed-loop success on matched task-initialization pairs, per-task and per-init success rates, paired repair/regression counts, and an explicit distinction between fake-quant behavior and packed-kernel deployment.

7 Implications

Evaluation. VLA quantization should be evaluated as a closed-loop policy perturbation. Reporting only MSE or only aggregate success can be misleading. Paired rollout flips are a compact way to expose whether a method is a monotonic improvement or a redistribution of successes.

Calibration. Calibration should not only minimize average layerwise or action-level error. It should consider which errors matter for downstream control. A useful calibration set should cover fragile contact states, low-margin success boundaries, and task slices where the FP16 policy is already weak. A control-aware calibration objective would weight errors by estimated closed-loop sensitivity, not only by their open-loop magnitude.

Fine-tuning. Quantization-aware fine-tuning is policy adaptation under low-precision constraints. If the objective is only behavior cloning on static observations, it may still miss closed-loop state distribution shift. More policy-aware objectives may be needed, including rollout-informed calibration, DAgger-style data aggregation, or task-slice reweighting.

Deployment claims. Our results are behavior-level evidence for fake W4A8 quantization. They do not establish latency, memory, or energy improvements. A complete deployment claim requires packed low-precision kernels and end-to-end inference profiling.

8 Limitations

This study has several limitations. The quantized implementation uses fake quantization, not a final packed int4/int8 deployment backend. Results are from one GR00T N1.5 checkpoint and LIBERO-10, not multiple VLA families. The main closed-loop benchmark has 150 task-initialization pairs; larger runs would provide tighter statistical confidence, and the current aggregate success intervals overlap. We do not run real-robot experiments. ATM/OHB are implemented as reproduction-oriented calibration mechanisms; exact implementation details may differ from the original paper. We evaluate success-rate flips, but do not yet provide trajectory-level geometric diagnostics for each flip.

These limitations do not weaken the central methodological point: local action approximation and closed-loop policy behavior are distinct evaluation targets.

9 Conclusion

Post-training quantization of VLA policies should be viewed as policy perturbation. Even small action errors can matter because they enter a feedback loop, alter the closed-loop state distribution, and interact with task-dependent success margins. In our GR00T/LIBERO reproduction, selective W4A8 quantization remains behaviorally robust and can shift aggregate point estimates, but those shifts come from task-dependent repair/regression redistribution rather than uniform dominance. Attention compensation methods reduce mean offline drift, yet their closed-loop effects are non-monotonic.

The main lesson is practical: VLA quantization papers should report paired closed-loop rollouts, not only offline drift or aggregate success. Small action errors matter when they change the trajectory basin.

A Detailed Experimental Tables

A.1 Standard 5-Trial FP16 and ATM+OHB W4A8

Table 4: Standard 5-trial LIBERO-10 comparison.

Task id	FP16	W4A8	ATM+OHB	Delta
0	5/5		5/5	0
1	3/5		4/5	+1
2	3/5		5/5	+2
3	5/5		5/5	0
4	4/5		4/5	0
5	5/5		5/5	0
6	3/5		2/5	-1
7	4/5		3/5	-1
8	1/5		0/5	-1
9	5/5		5/5	0
Total	38/50		38/50	0

A.2 Disjoint Init 5–14 FP16 and ATM+OHB W4A8

Table 5: Disjoint init 5–14 comparison.

Task id	FP16	W4A8	ATM+OHB	Delta
0	8/10		8/10	0
1	8/10		10/10	+2
2	9/10		8/10	-1
3	10/10		10/10	0
4	4/10		8/10	+4
5	10/10		10/10	0
6	3/10		5/10	+2
7	7/10		7/10	0
8	2/10		3/10	+1
9	9/10		7/10	-2
Total	70/100		76/100	+6

A.3 Ablation: W4A8 None, ATM, OHB

Table 6: Ablation over init 0–14.

Task id	None	ATM	OHB	ATM-None	OHB-None
0	10/15	11/15	13/15	+1	+3
1	13/15	13/15	14/15	0	+1
2	13/15	14/15	15/15	+1	+2
3	15/15	13/15	13/15	-2	-2
4	8/15	13/15	12/15	+5	+4
5	14/15	14/15	15/15	0	+1
6	9/15	10/15	8/15	+1	-1
7	8/15	8/15	8/15	0	0
8	9/15	4/15	6/15	-5	-3
9	14/15	14/15	12/15	0	-2
Total	113/150	114/150	116/150	+1	+3

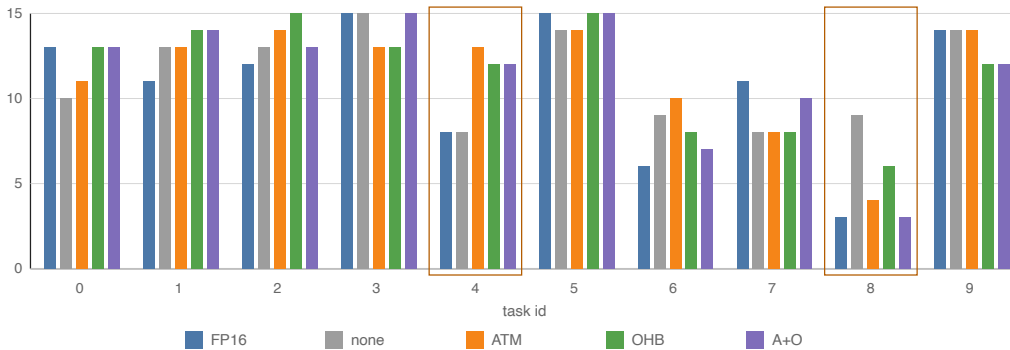


Figure 4: Task-wise success redistribution across policy variants. Aggregate changes arise from heterogeneous task-level gains and regressions rather than uniform improvement.

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